ON THE CORRELATION OF THE RATE OF HEART BEAT, BREATHING, BODILY MOVEMENT AND SENSORY STIMULI. By WALTER M. COLEMAN.

My first observations on this subject were made on animals in the Zoological Gardens, Regent's Park. The heart beats can be counted in various ways. It may be visible in the chest wall as in the squirrel, cheetah, leopard and seal; it may be shown by the vibrations of the whiskers; and sometimes, while the head is held up, the carotid pulse is visible.

The ratio of heart beat to the number of steps taken by the animals in quiet movement and to their ordinary quiet respiration was determined. It was found that usually the rate of heart beat was equal to, or an even multiple of, the rate of movement and breathing. The following are some of the observations made.

Animals	Duration of observations in seconds	Number of heart beats	Steps taken in walking	Ratio of breathing to heart beats
Civet ...	30	56 (seen on side)	56	1 to 4, 5, 4, 4, 4, 5
Cheetah ...	30	60 ,, ,,	60	1 to 5, 5, 6, 6, 6, 6, 4, 5
Leopard ...	30	80 (whiskers)	80	1 to 4, 4, 4, 4, 5

In 50 out of 300 observations the rate of steps and presumably of heart beat continually varied. But in nearly all these cases there was some obvious disturbing factor, such as the animals waiting to be fed, or a noise from the attendant. A bear walked at 10 steps in 6 seconds; teased by a boy, the steps were 10 in 6·2, 6·6, 5·8, 6·2, 6·6, 6·6, 6·6, 6·6, showing a recovery of steadiness in two minutes. The observations show, I think, that in animals when undisturbed there is a correlation between the discharge of nervous impulses causing movement, and the nervous discharge regulating the rate of the heart beat.

In order to test the matter further I have made observations in different ways on man.

(1) The subject was told to take his radial pulse with elbows held out, and, while taking the pulse, to walk with an easy swing at steady

pace paying more attention to his footsteps but continuing to observe his pulse. When the steps were faster than the pulse it was found that the pulse rate increased, and at rates within the limits of 90 to 120 a minute the pulse rate became that of the steps. If the test is repeated with the spine held stiff and the attention concentrated instead of easy, the pulse and steps fail to coincide.

(2) In the next test the subject's pulse was taken and he was then asked to walk in an indifferent mood. He was not told the object of the test. The rate of his steps was taken. 14 out of 24 subjects stepped at the rate taken for their pulse. As with several of the tests, it is best not to make this test twice on the same subject; for if he suspects that the influence of his heart on his footsteps is being tested, self-consciousness may become a source of error.

(3) The effect of attention to periodic sounds was also tried. The subject's pulse rate was taken for 15 seconds. The subject sat erect and poised and a metronome was placed before him and set beating at a rate 15–30 faster, or 10–15 slower, than his pulse. He was asked to give his undivided attention to the sounds. After half a dozen beats, his pulse was again taken for 15 seconds. It was found in nine cases out of ten that the pulse rate had become that of the metronome; thus in different observations the rate changed from 60 to 72, from 64 to 84, from 80 to 108, and from 56 to 48. The subject is not to be told that his pulse is expected to change its rate; and the clicking sound is not allowed beforehand lest his ears become tired. The attention lapsed or the mind wandered in some cases in less than half a minute. Those with power of sustained attention can keep the pulse in accord with the clicks for several minutes. With most subjects there was a perceptible swing of head to right and left. If tracings of the pulse changes are taken, the self conscious state induced becomes a source of error, especially if a sphygmograph is used with the uncomfortably high pressure necessary for air-transmission and the body held still to prevent risk of spoiling the record. The pocket Dudgeon sphygmograph used at its lightest pressure does not seem to interfere with the response; the records show the transition to a new rate to take place within 2 or 3 heart beats.

(4) Subjects singing with others were asked to pause in the song and test the pulse. It was found as a rule to keep time with the music.

(5) In a number of instances, while listening to a fluent or impressive speaker an auditor tested his own pulse and found a pulse-beat to fall with each accented syllable of the speaker's words.

(6) The rate of the heart may also be brought into time with move-

ment without the aid of impulses from the volitional centre by passively swinging the arm or leg of the subject at regular intervals.

That sudden changes in intra-thoracic pressure from sudden opening and closing of the glottis may be a factor in controlling the heart is indicated by the two following tests; which may also explain why many persons cannot take their own pulse without changing its rate:

The pulse of one person is taken by another and then by himself; if the rates differ he takes his own pulse again, this time only counting it mentally, keeping the throat relaxed and open. The pulse in most cases will now agree with the rate found by the first observer. The second test is to take the pulse in the relaxed way just described, and then to take it counting aloud or in a forcible whisper, calling each count a little ahead of the pulse beat. Its rate will thereby be quickened 10 or 15 p.c. Then it is taken with the count slightly dragging after the beats; this retards the rate 10 or 15 p.c. Since the counting is continually accelerated or retarded the pulse is not allowed to catch up with the counting.

Observation of the bare abdominal wall showed that for each act of attention, whether sensory or motor, there occurred a slight twitch or pause in the respiratory movement of the wall. Thus the muscular wall of the body, acting with the diaphragm, and perhaps aided by the "stroke of the glottis," probably gave mechanical stimuli to the heart, bringing it to the new rate. If the subject sat in a collapsed posture, instead of erect and poised, the pulse-rate was not controlled by attention to periodic stimuli. Since the pericardium is bound down to the diaphragm, each respiratory movement acts directly upon it.

Besides the sudden dynamic changes in pressure upon the heart just mentioned, its rate is probably influenced by the mechanical effects of the footsteps and other intermittent movements; for it is found that if all interruptions and jarring are avoided, as in using a bicycle ergometer where the rhythmic changes in exertion are very gradual, the movements fail to impart their rate to the heart. If by cutting out sudden periodic changes the effect ceases it is probable that they are necessary to the effect.

I have several times pointed out that stiffness interferes with the response of the heart to rhythm; this effect is probably likewise due to the stiffness preventing intermittence (in exertion and relaxation). The stiffness may be occasioned by self-consciousness, by emotion, or by the attachment to the body of delicate apparatus, the use of which requires immobility. The heart seems to take up a new rate best when the body and head are free to swing from side to side. For anatomical

reasons the pendulum effect of a right and left swing is more regular than the effect of a swing back and forth.

The experiments seem to show that voluntary attention involves muscular activity. Yet it appears that mere voluntary attention alone, which does not also include intermittent, periodic action of the respiratory muscles, will not bring the heart to the rate of the stimulus. However, the stimulus may disturb the heart and change its rate, for doubtless, in all voluntary attention, impulses irradiate directly to the heart by way of the vagus. Though there is doubtless associated activity of this mechanism, the response of the heart to external periodic stimuli appears to be not so much direct as mediated through the rhythmic action of the glottis, abdominal wall and swings of limb or trunk, though the last may be so slight as to pass unperceived.

These bodily responses are prevented by rigidity; and this in turn has its source in strain from unsymmetrical posture, or in pain or discomfort or in the immobility, for instance, necessary during observations with delicate apparatus where any motion of the subject would spoil the record. Stiffness may also arise in the self-consciousness of tests made before spectators. Any emotional, asphyxial or apnœic condition which necessitates irregular breathing will obviously interfere with the regular action of glottis, body wall and the rhythmic swing.

In man as in the lower animals, as has already been shown, irregular rhythm of body and probably also of heart forms a part of emotional reaction. This tends to prevent the taking up of regular rhythm; for I find that a heart that is already beating at a regular rate will fall in with periodic stimuli of a new rate when an irregularly beating heart remains unaffected.

Metabolic demands upon the breathing cannot, of course, set the heart at a definite rate, but those demands must not place a hindrance in the way of the respiratory response to periodic stimuli. For instance in reducing the pulse rate by attention to the slow beats of a metronome the heart did not slow its rate unless the breathing was relaxed and the sounds counted in a drone or drawl in a half indifferent mood. Experimental evidence of this is found if the test of the heart for periodic response is combined with the writer's test (this *Journal*, **53**, p. 361) for ventilation of the blood: it will be found that the heart takes new rhythms best when ventilation is balanced, this being known by the succession of paired and unpaired images. (For this test the subject must be normal, not one who gets a fixed image, nor one—usually over 60 years old—who, because of conjugate deviations, gets no image.)

The problem has been, not to study the well-known responsiveness of the heart to almost all stimuli, but to find under what conditions, if any, the heart will keep in time with perfectly periodic stimuli.

Summary.

1. Accord of rates is usual between heart, footsteps and breathing.

2. The heart will take up the rate of periodic movements or sensory stimuli within limits (about 15 p.c. below or 30 p.c. above its rate) if the body is free of stiffness, in erect, easy poise, and the attention steady but flexible.

3. The readjustment begins at once and is completed in about three heart beats.

4. Emotion, strong exertion, or unusual metabolic demands may prevent the accord of rates.

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